A Modest Proposal for Operationalising Dramatic Texts

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Operationalisation in computational drama studies

As evidenced by recent scholarly debate (see e.g. Krautter 2022, Pichler and Reiter 2022), operationalisation remains a core task in any computational approach to literature, and its importance for a robust research pipeline cannot be underestimated. "[B]uilding a bridge [...] from the concepts of literary theory, through a form of quantification, to literary texts" (Moretti, 2013b, 1), operationalisation is a delicate procedure that requires a balancing act between the specificity of the literary text and the algorithmic requirements of the machine. In the context of computational drama analysis – the subfield of computational literary studies that "attempts to model, analyse and interpret dramatic texts using computational methods" (Andresen and Reiter, 2024b, 1) – one could say, with some approximation, that operationalisation practices coincide with the identification and quantification of some crucial dramatic features and the modelling of their relations.

However, defining the 'key components' of a play has always been a complex theoretical problem in itself; the debate began already with Aristotle's comments on the six constituent parts of tragedies (*Poetics* 1450a, 5-10). Currently, many scholars seem to have settled on working with what Kretz (2012) calls the three 'elementary components' of drama: characters, dialogue and plot. Interestingly, a similar division was already introduced by Russian formalist Boris Yarkho, who spoke of three distinctive elements in theatrical texts: 'dialogue' (диалог), 'characters' (персонажи), and 'action' (действие) (Yarkho, 2006, 410).

With some approximation, it can be argued that these high-level components manifest themselves in different textual aspects (content, style, structure...), which in turn can be subjected to different operationalisation practices and explored using different methods. As singled out by Willand et al. (2017), three main approaches seem now prevalent in the field of computational drama analysis: topic modelling of dramatic genres, exploration of network structures, and analysis of character speech.

Measurements and metrics

At a concrete level, operationalisation begins with the choice of the more appropriate features to describe a phenomenon. Most scholars attempting to model dramatic plays have therefore collected readily available metrics (e.g. the number of characters in a play) or developed new measures to capture what they considered key aspects of texts (e.g. the protagonist's features). This second approach should be not considered, however, a goal in itself, but should rather allow to gain a hermeneutical advantage over existing metrics and/or competing non-computational methods. Furthermore, it should be kept in mind that "[m]easures can be taken on any quantifiable aspect of a text, but finding out the significance of that metric for understanding the text, or better, mapping that metric onto a pre-existing critical concept (such as style, plot, or theme), is crucial to making sense of what is being measured" (Algee-Hewitt, 2017, 759).

To revisit some earlier examples of approached for defining metrics in drama, Yarkho's (2006) study on Corneille was based on fifteen features he considered capable of distinguishing comedies from tragedies, while his investigation of speech distribution in five-act tragedies relied on four character-related measures. Interestingly, his quantitatively oriented, but still 'analogical' approach allowed him to create also highly specific features, such as 'saturation' – a compound metric indicating, among other things, 'the percentage of characters acting in their own interest' (Gasparov, 2017, 141-42). Other Russian (neo)formalists followed Yarkho's lead and developed their own metrics. Revzina and Revzin (1971), for example, introduced measures such as 'character connectedness', 'dramatic engagement rank', and 'probabilistic link', which anticipated several later DH efforts; their approach was later reprised by Sapogov (1974). Sperantov (1998), for his part, explored stage directions in Russian verse tragedy through measures such as frequency, average length, presence within verses, lexical diversity, etc.

Similarly, contemporary scholars have kept developing new metrics tailored to their research needs. For example, Trilcke et al. (2017) have taken inspiration from early work by Xanthos et al. (2016) to address the progressive structuring of

plot. To this aim, they computed several 'event-based' (i.e., “identify[ing] or characteris[ing] a particular point in time within a drama”), and 'progression-based' (i.e., “allow[ing] a general characterisation of the transformation of a dramatic network”) measures (Fischer et al., 2017, 2-3). In his quantitative study of English early modern drama, Algee-Hewitt (2017) has instead built on standard network measures of centrality to create two advanced metrics – 'protagonism' and 'mediatedness'.

Vectorisation and its epistemological implications

The previous examples show how scholars often focused on the measurement and comparison of little sets of metrics. However, a deeper comprehension of literary history and cultural evolution could arguably be achieved only by relying on a broader pool of features, covering as many aspects of a play as possible – not only network structures, but also textual size, speech distribution, plot development, etc.

Nonetheless, dealing with multiple features at once is challenging both from a computational and a conceptual point of view. To tackle this problem, one possible approach might be collecting metrics in ordered lists, i.e., in mathematical vectors. So far, the most common use case for vectors in DH research is word embeddings, which aim to capture the meaning of words based on their distributional properties (cf. Firth 1957). [1] Several approaches have been developed to map words to real-valued vectors, ranging from static vector space models and latent semantic analysis to neural networks-based methods, including transformers.

Within drama studies, a similar logic can be followed to construct *play embeddings* – vectors made up of measures that embody different textual aspects of the text. Such embeddings can be easily constructed e.g. by relying on information extracted by the API of the *DraCor* project (Fischer et al. 2019), which calculates a wide range of statistics for each play included in their dramatic corpora (Figure 1). These metrics, describing for instance the properties of character networks or speech distribution among figures, can be collected in vectors which effectively represent the ‘form’ of a pièce, and can be compared to other vectors to measure formal (dis)similarities between dramatic works.

	id	name	yearNormalized	size	averageClustering	density	averagePathLength	averageDegree	diameter
0	greek000026	aeschylus-agamemnon	-458.0	7	0.690476	0.523810	1.476190	3.142857	2
1	greek000004	aeschylus-eumenides	-458.0	7	0.771429	0.619048	1.133333	3.714286	2
2	greek000008	aeschylus-libation-bearers	-458.0	8	0.883333	0.750000	1.250000	5.250000	2
3	greek000022	aeschylus-persians	-472.0	5	0.600000	0.600000	1.400000	2.400000	2
4	greek000011	aeschylus-prometheus-bound	-444.0	7	0.823810	0.476190	1.523810	2.857143	2

Figure 1. A section of the metadata table generated by the DraCor API for the Greek corpus. The numeric values, where relevant, can be collected in vectors to approximately represent the plays’ ‘form’.

While the pipeline for producing play embeddings is relatively straightforward, one might nonetheless question the epistemic validity of the method itself. Compressing texts into a series of numbers that purport to represent some of their most salient aspects, and then comparing these 'bags of numbers' to make claims about literary history, could easily be seen as yet another example of the simultaneously ‘hubristic’ and ‘reductionist’ mindset often ascribed to the digital humanities.

It could be argued, however, that drama vectorisation is just another expression of the 'cultural technique of flattening' Krämer (2023a, 2023b) sees as fundamental not only to DH endeavours, but to humanistic inquiry as a whole. In her vision, the opposition between traditional modes of scholarly interpretation, represented by 'deep' hermeneutics, and data-driven, 'shallow' digital studies is indeed misconstrued, insofar as the humanities have always used 'superficial' representations of cultural objects as their research objects.

Consequently, computational methods seem to differ from traditional ones only in the extension of their 'flattening' procedure, i.e. in the way they transform cultural objects into machine-actionable data. From this perspective, drama vectorisation should not be interpreted as a reductionist attempt to overtly simplify the complexity of theatrical texts, but rather just as a continuation of formalist morphological thinking with modern tools.

Research applications

In general terms, vectorisation could hardly be defined as a novel approach, as many well-established methods in computational literary studies – such as stylometry and topic modelling – rely on transforming texts or words into vectors. This procedure also underpins for what Michael Gavin (2018: 641) calls “vector semantics”, i.e., a “sub-

field of computational linguistics that uses statistical measurements to model the meanings of words". Only recently, however, scholars have begun to explore vectorisation of texts based not on some semantic features, but on their formal characteristics. To wrap up this survey, we briefly review some new promising applications of this research paradigm.

Aside from the pioneering experiments of Solomon Marcus [2] and from the dissertation by Wendell (2021), which used vectorisation only where necessary to replicate some experiments by Yarkho, the first 'fully-fledged' application of vectorisation can be arguably found in Szemes and Vida (2024). Their study, aimed at distinguishing comedies from tragedies in German and Shakespearean drama, created play embeddings based on a set of standard network features and some derived metrics. While their attempt at classification found no major structural genre differences, they were still able to identify some properties predictive of generic alignment, as previously speculated by Ilsemann (1995).

Giovannini and Skorinkin (2024) repurposed this method to study a sample of German and French opera libretti, focusing on exploring their unique formal properties and evolution over time. While their results were limited, confirming well-known challenges in modelling dramatic texts using structural features alone, they did highlight some formal patterns typical of musical drama. Finally, Giovannini (2024) attempted the first comprehensive use of vectorisation to investigate the formal evolution of early modern European drama. By creating comprehensive vectors to represent plays from different traditions and applying several exploratory techniques, he managed to map the evolution of drama over a 150-years timeframe and provide an empirical test of some well-established critical theories.

Notes

1. For applications of word embedding models in DH context see e.g. Schöch (2022); some theoretical considerations in Dobson (2021).
2. For example, he created binary vectors for each character registering their presence or absence from the stage, thus creating a sort of Boolean matrix, and then calculated the difference between vectors (i.e., the difference in stage prominence among characters) via Hamming distance (Marcus 1970, ch. 7).

Bibliography

Algee-Hewitt, Mark (2017). "Distributed Character: Quantitative Models of the English Stage, 1550–1900". *New Literary History*, 48(4):751–782. DOI: https://doi.org/10.1353/nlh.2017.0038 .

Andresen, Melanie and Nils Reiter (2024). "Introduction". In Melanie Andresen and Nils Reiter, editors, *Computational Drama Analysis: Reflecting on Methods and Interpretations*, 1–6. De Gruyter, Berlin, Boston. DOI: https://doi.org/10.1515/9783111071824-001 .

Dobson, James E. (2021). "Vector hermeneutics: On the interpretation of vector space models of text". *Digital Scholarship in the Humanities*, 37(1):81–93. DOI: https://doi.og/10.1093/llc/fqab079 .

Firth, John R. (1957). "A Synopsis of Linguistic Theory, 1930–55". *Studies in Linguistic Analysis*, 23:1–32.

Fischer, Frank, Mathias Göbel, Dario Kampkaspar, Christopher Kittel, and Peer Trilcke (2017). "Network Dynamics, Plot Analysis: Approaching the Progressive Structuration of Literary Texts". In *DH2017 Book of Abstracts*, Montréal. McGill University. URL: https://dh2017.adho.org/abstracts/071/071.pdf .

Fischer, Frank, Ingo Börner, Mathias Göbel, Angelika Hechtl, Christopher Kittel, Carsten Milling, and Peer Trilcke (2019). "Programmable Corpora: Introducing DraCor, an Infrastructure for the Research on European Drama". In *DH 2019 Book of Abstracts*, Utrecht. University of Utrecht. DOI: https://doi.org/10.5281/zenodo.4284002 .

Gasparov, Mikhail (2017). "Boris Yarkho's Works on Literary Theory". *Studia Metrica et Poetica*, 3(2):130–150. DOI: https://doi.org/10.12697/smp.2016.3.2.05 .

Gavin, Michael (2018). "Vector semantics, William Empson, and the study of ambiguity". *Critical Inquiry*, 44(4): 641-673. URL: https://www.jstor.org/stable/26547856 .

Giovannini, Luca (2024). "Evolutive Dynamics in Early Modern European Drama: A Computational Approach". PhD thesis, University of Potsdam and University of Padova.

Giovannini, Luca, and Daniil Skorinkin (2024). "Computational Approaches to Opera Libretti". *Journal of Computational Literary Studies*, 3. DOI: https://doi.org/10.48694/jcls.3595 .

Ilsemann, Hartmut (1995). "Computerized Drama Analysis". *Literary and Linguistic Computing*, 10(1):11–21. DOI: 10.1093/llc/10.1.11. URL: http://dx.doi.org/10.1093/llc/10.1.11 .

Krautter, Benjamin (2022). "Die Operationalisierung als Interdisziplinäre Schnittstelle der Digital

Humanities". *Scientia Poetica*, 26(1):215–244. DOI: https://doi.org/10.1515/scipo-2022-009 .

Kretz, Nicolette (2012). "Grundelemente (1): Bausteine des Dramas (Figur, Handlung, Dialog)". In Peter W. Marx, editor, *Handbuch Drama*, 105–121. J.B. Metzler, Stuttgart. DOI: https://doi.org/10.1007/978-3-476-00512-0_9 .

Krämer, Sybille (2023a). "Should We Really 'Hermeneutise' the Digital Humanities? A Plea for the Epistemic Productivity of a 'Cultural Technique of Flattening' in the Humanities". *Journal of Cultural Analytics*, 7(4). DOI: https://doi.org/10.22148/001c.55592 .

Krämer, Sybille (2023b). "The Cultural Technique of Flattening". *Metode*, 1. URL: https://metode.r-o-m.no/en/articles/essay/the-cultural-technique-of-flattening .

Marcus, Solomon (1970). *Poetica matematica*. Academiei, Bucuresti.

Moretti, Franco (2013). "'Operationalizing': or, the Function of Measurement in Modern Literary Theory", December. URL: https://litlab.stanford.edu/LiteraryLabPamphlet6.pdf .

Pichler, Axel, and Nils Reiter (2022). "From Concepts to Texts and Back: Operationalization as a Core Activity of Digital Humanities". *Journal of Cultural Analytics*, 7(4). DOI: 10.22148/001c.57195. URL: https://culturalanalytics.org/article/57195-from-concepts-to-texts-and-back-operationalization-as-a-core-activity-o .

Revzina, Olga G., and Isaak I. Revzin (1971). "Nekotorye Matematicheskie Metody Analiza Dramaturgicheskogo Postroeniia" ["Some Mathematical Methods for Analyzing Dramatic Constructions"]. *Tochnye Metody v Issledovaniakh Kul'tury i Iskusstva (Materialy k Simpoziumy)*, 291–300.

Sapogov, Vyacheslav A. (1974). "Nekotorye Harakteristiki Dramaturgicheskogo Postroeniia Komedii A. N. Ostrovskogo »Les«" ["Some Characteristics of the Dramatic Construction of A. N. Ostrovsky's Comedy »The Forest«"]. In Vyacheslav Alexandrovich Sapogov, editor, *A. N. Ostrovskij i Russkaia Literatura* [A. N. Ostrovsky and Russian Literature], 60–69. Yaroslavl State Pedagogical Institute, Kostroma.

Schöch, Christof (2022). "Quantitative Semantik. Word Embedding Models für Literaturwissenschaftliche Fragestellungen". In Fotis Jannidis, editor, *Digitale Literaturwissenschaft*, 535–562. J.B. Metzler, Stuttgart. DOI: https://doi.org/10.1007/978-3-476-05886-7_22 .

Sperantov, Vladimir V. (1998). "Poetika Remarki v Russkoy Tragedii XVIII - Nachala XIX vv. (K Tipologii Literaturnykh Napravleniy)" [The Poetics of Stage Directions in Russian Tragedy of the Eighteenth and Early Nineteenth Centuries (On the Typology of Literary Movements)]. *Philologica*, 5(11/13): 9–48. URL: https://rvb.ru/philologica/05/05sperantov.htm .

Szemes, Botond, and Bence Vida (2024). "Tragic and Comical Networks: Clustering Dramatic Genres According to Structural Properties". In Melanie Andresen and Nils Reiter, editors, *Computational Drama Analysis: Reflecting on Methods and Interpretations*. De Gruyter, Berlin; Boston. URL: https://doi.org/10.1515/9783111071824-009 .

Trilcke, Peer, Frank Fischer, Mathias Göbel, Dario Kampkaspar, and Christopher Kittel (2017). "Netzwerkdynamik, Plotanalyse – Zur Visualisierung und Berechnung der Progressiven Strukturierung Literarischer Texte". In *DHd2017 Book of Abstracts.*, Bern. DOI: https://doi.org/10.5281/ZENODO.4622799 .

Wendell, Inna A. (2021). *A Statistical Analysis of Genre Dynamics: Evolution of the Russian Five-Act Comedy in Verse in the Eighteenth and Nineteenth Centuries*. PhD thesis, University of California, Los Angeles. URL: https://escholarship.org/uc/item/9rr5k9p7 .

Willand, Marcus, Peer Trilcke, Christof Schöch, Nanette Rißler-Pipka, Nils Reiter, and Frank Fischer (2017). "Aktuelle Herausforderungen der Digitalen Dramenanalyse". In *DHd2017 Book of Abstracts*, Bern. DOI: https://doi.org/10.5281/ZENODO.4622643 .

Xanthos, Aris, Isaac Pante, Yannick Rochat, and Martin Grandjean (2016). "Visualising the Dynamics of Character Networks". In *DH 2016 Book of Abstracts*. URL: http://dh2016.adho.org/abstracts/407 .

Yarkho, Boris (2006). *Metodologiia Tochnogo Literaturovedeniia: Izbrannye Trudy po Teorii Literatury* [Methodology for the Exact Study of Literature: Selected Works in Literary Theory]. Jazyki Slavianskikh Kul'tur, Moskva.